



The low-energy ${}^7\text{Be}(p, \gamma){}^8\text{B}$ cross section from an R-matrix approach

F.C. Barker

*Department of Theoretical Physics, Research School of Physical Sciences and Engineering,
Australian National University, Canberra, ACT 0200, Australia*

Received 19 December 1994

Abstract

Calculation of the low-energy ${}^7\text{Be}(p, \gamma){}^8\text{B}$ cross section, using R-matrix formulae with parameter values chosen to fit experimental data for the mirror reaction ${}^7\text{Li}(n, \gamma){}^8\text{Li}$, leads to a ‘low’ value of the S factor, $S_{17}(0) = 17 \pm 3 \text{ eV} \cdot \text{b}$. This value agrees with those obtained from potential-model calculations, from calculations of nuclear-vertex constants, and from a Coulomb-dissociation experiment, but is less than the values obtained from cluster-model calculations.

1. Introduction

Knowledge of the low-energy ${}^7\text{Be}(p, \gamma){}^8\text{B}$ cross section is of great importance in the solar-neutrino problem [1]. The cross section is usually represented by the zero-energy value of the astrophysical S factor, $S_{17}(0)$. Over the years, the value of $S_{17}(0)$ adopted in the standard-solar-model calculations of Bahcall and his collaborators has decreased from $43 \text{ eV} \cdot \text{b}$ [2] to $22.4 \text{ eV} \cdot \text{b}$ [3], though the assigned uncertainty has remained at about 10%. The smaller the value of $S_{17}(0)$, the smaller the calculated flux of high-energy solar neutrinos and the smaller the discrepancy with measured values.

The ${}^7\text{Be}(p, \gamma){}^8\text{B}$ cross section has been measured at laboratory energies ($E_p \gtrsim 130 \text{ keV}$) and extrapolated to low energies in order to determine $S_{17}(0)$; uncertainties in the measurements and in the extrapolation procedures have been discussed by Barker and Spear [4], who suggested a reduction in the then-adopted value of $S_{17}(0)$ by about 30% to $S_{17}(0) \approx 17 \text{ eV} \cdot \text{b}$. Kim et al. [5] went further and claimed a ‘solution’ of the solar-neutrino problem by using a different extrapolation procedure that led to a much smaller low-energy cross section.

From Coulomb dissociation of ${}^8\text{B}$, Motobayashi et al. [6] obtained $S_{17}(0) = 16.7 \pm 3.2$

eV · b. Langanke and Shoppa [7] suggested that this value should be reduced by about 25% due to E2 contributions, but this has been refuted [6,8].

Calculation of the ${}^7\text{Be}(p, \gamma){}^8\text{B}$ cross section using a direct-capture potential model, with parameter values determined by fitting measured properties of the mirror reaction ${}^7\text{Li}(n, \gamma){}^8\text{Li}$, was suggested by Tombrello [9] and developed by Barker [10], who found $S_{17}(0)$ values in the range 13.5 to 22.3 eV · b. A potential-model calculation by Kim et al. [11], which gave $S_{17}(E_{\text{c.m.}} = 20 \text{ keV}) = 24 \text{ eV} \cdot \text{b}$ corresponding to $S_{17}(0) = 24.7 \text{ eV} \cdot \text{b}$, was criticized by Barker [12]; similar criticisms apply to the calculation by Krauss et al. [13], which gave $S_{17}(0) = 23.6\text{--}24.9 \text{ eV} \cdot \text{b}$.

Cluster-model calculations by Descouvemont and Baye have given $S_{17}(0) = 30 \text{ eV} \cdot \text{b}$ [14] and $27\text{--}30 \text{ eV} \cdot \text{b}$ [15], and by Langanke and collaborators $25.4 \text{ eV} \cdot \text{b}$ [16], $22.4 \pm 2.1 \text{ eV} \cdot \text{b}$ [17] and $25\text{--}27 \text{ eV} \cdot \text{b}$ ($S_{17}(E_{\text{c.m.}} = 20 \text{ keV}) = 24.6\text{--}26.1 \text{ eV} \cdot \text{b}$) [18].

Mukhamedzhanov and Timofeyuk [19] have pointed out that $S_{17}(0)$ is determined solely by C_{1s} , the asymptotic normalization constants of the ${}^8\text{B}$ ground-state wave function in the ${}^7\text{Be}(\text{g.s.}) + p$ channel, and these are closely related to the nuclear-vertex constants G_{1s} . Values of G_{1s} obtained in a microscopic calculation [20], using an NN potential that fits measured values of G_{1s} for other p-shell nuclei, led to $S_{17}(0) = 15.5 \text{ eV} \cdot \text{b}$ [19], which has recently been updated to $17.6 \text{ eV} \cdot \text{b}$ [21]. Riisager and Jensen [22] have discussed other ways in which information might be obtained on the proton tail of the ${}^8\text{B}$ wave function, and so on $S_{17}(0)$, and have suggested $S_{17}(0) = 12\text{--}20 \text{ eV} \cdot \text{b}$.

It therefore seems that a variety of approaches indicates a value of $S_{17}(0)$ smaller than the presently-adopted value [3,17], whereas the cluster-model calculations mostly favour a larger value.

Here we investigate an approach that has not previously been used for this problem. This is to calculate the ${}^7\text{Be}(p, \gamma){}^8\text{B}$ cross section using formulae from R-matrix theory [23], extended for photon reactions by the inclusion of channel contributions [24]. The R-matrix parameters are assumed to be the same as for the mirror reaction ${}^7\text{Li}(n, \gamma){}^8\text{Li}$, and are determined by fitting ${}^7\text{Li} + n$ data. Previously [10], the potential-model approach was used to calculate only the so-called nonresonant contributions to the ${}^7\text{Li}(n, \gamma){}^8\text{Li}$ and ${}^7\text{Be}(p, \gamma){}^8\text{B}$ cross sections, which were attributed to direct capture of s- and d-wave nucleons with E1 transitions to final p-states. Narrow peaks observed in the cross sections at $E_n \approx 0.25 \text{ MeV}$ and $E_p \approx 0.72 \text{ MeV}$ respectively (see Figs. 1 and 2 below), were attributed to capture of p-wave nucleons into 3^+ or 1^+ excited states of the compound nucleus, followed by M1 transitions to the 2^+ ground states; R-matrix parameter values required to fit the data were compared with those obtained from shell-model calculations [10]. Here, the E1-capture contribution, which is the part that determines $S_{17}(0)$, is also treated using R-matrix theory. Because the negative-parity levels of the compound nucleus are very broad, they do not produce identifiable peaks in the cross sections.

A large body of ${}^7\text{Li} + n$ data has already been fitted with multilevel, multichannel R-matrix formulae by Knox et al. [25], who did not, however, include the ${}^7\text{Li}(n, \gamma){}^8\text{Li}$ cross section. Knox et al. initially obtained values of the level energies and reduced width amplitudes from a shell-model calculation, and found that only a few of these

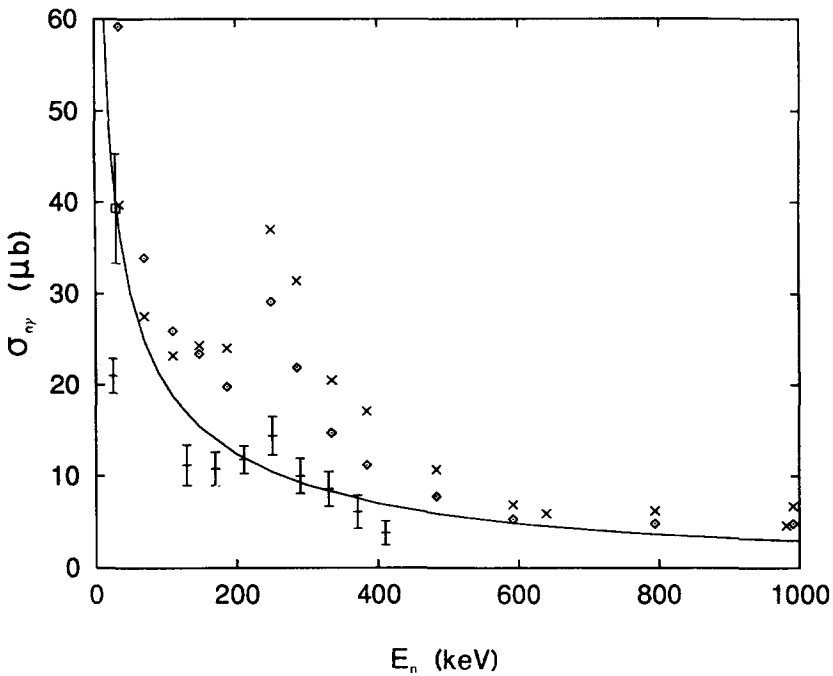


Fig. 1. Cross section $\sigma_{n\gamma}$ for ${}^7\text{Li}(n, \gamma){}^8\text{Li}$ as a function of neutron energy E_n . The experimental data are from Imhof et al. [31] (crosses and diamonds), Wiescher et al. [33] (pluses), and Nagai et al. [34] (square). The curve is calculated using the first row of modified values in Table 3 and excludes any contribution from the resonance at $E_n \approx 250$ keV.

values needed to be changed in order to fit the data. The ${}^7\text{Li}(n, \gamma){}^8\text{Li}$ cross section can be expressed in terms of these same parameters, as well as the internal contributions to the photon reduced-width amplitudes. Other useful data for quantities that depend on the same parameter values are values of the spectroscopic factors for breakup of the low-lying ${}^8\text{Li}$ levels into ${}^7\text{Li}(\text{g.s.}) + n$, obtained by analysis of the stripping reaction ${}^7\text{Li}(\text{d}, \text{p}){}^8\text{Li}$.

This R-matrix approach is not necessarily better than those used previously but it is different, and it may be of interest to see if it favours a ‘low’ or ‘high’ value of $S_{17}(0)$.

The next section gives the experimental data, mainly for the ${}^7\text{Li} + n$ system, and the relevant R-matrix formulae are in Section 3. Section 4 gives the procedures and results, and finally we discuss these and previous results.

2. Experimental data

The scattering lengths a_s (s = channel spin) for elastic scattering of neutrons on ${}^7\text{Li}$ are [26]

$$\begin{aligned} a_1 &= 0.87 \pm 0.07 \text{ fm}, \\ a_2 &= -3.63 \pm 0.05 \text{ fm}. \end{aligned} \tag{1}$$

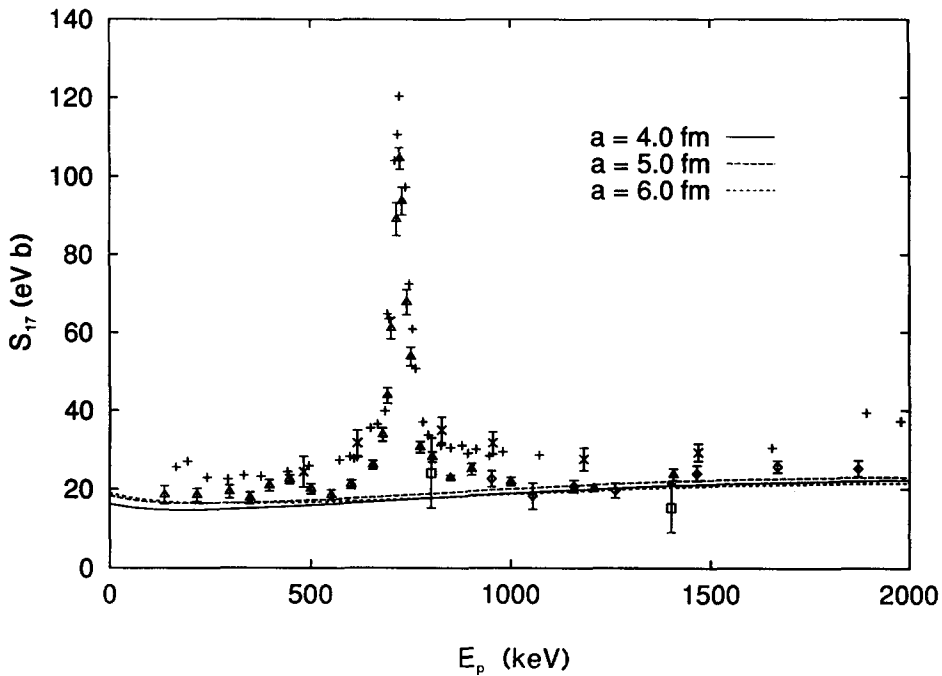


Fig. 2. S factor for ${}^7\text{Be}(p, \gamma){}^8\text{B}$ as a function of proton energy E_p . The experimental data are from Kavanagh [45] (squares), Parker [46] (crosses), Kavanagh et al. [47] (pluses), Vaughn et al. [48] (diamonds) and Filippone et al. [37] (triangles). The calculated curves are for the indicated values of the channel radius and exclude any contribution from the resonance at $E_p \approx 720$ keV.

From the radiative capture of thermal neutrons by ${}^7\text{Li}$, the total cross section is [27]

$$\sigma_{n\gamma}(E_{\text{th}}) = 45.4 \pm 3.0 \text{ mb}, \quad (2)$$

the branching ratio for capture to the 1^+ first-excited state of ${}^8\text{Li}$ is [27]

$$\mathcal{R} = 0.106 \pm 0.010, \quad (3)$$

and a lower limit has been placed on the fraction W_+ of the capture that proceeds via the initial channel spin $s = 2$ [28]:

$$W_+ \geq 0.86. \quad (4)$$

Earlier less-precise values of $\sigma_{n\gamma}(E_{\text{th}})$ were 33 ± 5 mb [29], 42 ± 10 mb [30], 40 ± 8 mb [31], and 40 ± 12 mb [32].

Various groups have measured the ${}^7\text{Li}(n, \gamma){}^8\text{Li}$ cross section at higher neutron energies. Imhof et al. [31] measured $\sigma_{n\gamma}$ for $E_n = 40$ – 1000 keV, which includes the peak at $E_n \approx 250$ keV, using the ${}^6\text{Li}(n, t){}^4\text{He}$ and ${}^{127}\text{I}(n, \gamma){}^{128}\text{I}$ cross sections for normalization. These gave, for example, $\sigma_{n\gamma}(E_n = 40 \text{ keV}) = 39$ and $59 \mu\text{b}$ respectively. These reference cross sections have since been modified. Wiescher et al. [33] covered the range $E_n = 25$ – 420 keV, and obtained cross sections that were about one half those of Imhof et al. In particular, they obtained $\sigma_{n\gamma}(E_n = 25 \text{ keV}) = 21.0 \pm 1.9 \mu\text{b}$. Both of these groups [31,33] used activation measurements, so that their cross-sections

Table 1

Experimental values of widths and spectroscopic factors for low-lying levels of ^8Li and ^8B

J^π	Γ (keV)		$\mathcal{S}$ ^8Li
	^8Li	^8B	
2^+			$0.88^a, 0.87^b$
1^+		37 ± 5^c	$0.47^a, 0.48^b$
3^+	33 ± 6^d	390 ± 40^e	0.25^a

^a Macfarlane and French [35], $^7\text{Li(d,p)}^8\text{Li}$, PWBA, $\theta_0^2 = 0.06$.^b Schiffer et al. [36], $^7\text{Li(d,p)}^8\text{Li}$, DWBA.^c Filippone et al. [37].^d Ajzenberg–Selove [38].^e McGrath et al. [39].

tion measurements include decays to both the ^8Li ground state and first-excited state. Nagai et al. [34] detected the decay γ -rays and obtained for the ground-state decay $\sigma_{n\gamma_0}$ ($E_n = 30$ keV) = 35.4 ± 6.0 μb .

The $^7\text{Li(n, } \gamma)^8\text{Li}$ cross section is expected to be proportional to $1/v$ (v = neutron velocity) at least over the energy range from thermal energies to $E_n = 40$ keV; from $\sigma_{n\gamma}(E_{\text{th}}) = 45.4$ mb and $\mathcal{S} = 0.106$ given above, one would then expect $\sigma_{n\gamma}(E_n = 40$ keV) ≈ 36 μb , $\sigma_{n\gamma}(E_n = 25$ keV) ≈ 45 μb and $\sigma_{n\gamma_0}(E_n = 30$ keV) ≈ 37 μb . Thus, relative to the thermal-neutron values, the cross-section values of Imhof et al. are somewhat high, those of Wiescher et al. are too low by a factor of about two, and those of Nagai et al. are in good agreement.

Experimental values of the widths of the low-lying positive-parity levels of ^8Li and ^8B , and of the spectroscopic factors of the ^8Li levels for the $^7\text{Li(g.s.)} + \text{p-wave}$ neutron channel, are given in Table 1. In our fits we use the $\mathcal{S}$ values of Schiffer et al. [36], with an assumed uncertainty of 20% (from adding in quadrature uncertainties of 15% for absolute cross sections and 15% for variations in formulae [36]).

3. R-matrix formulae

Barker and Kajino [24] have given R-matrix formulae for the cross section for a radiative-capture reaction, for the general case of electric multipole radiation. These formulae include channel contributions, which are expected to be specially important in the present work because of the low binding energies of the final states. Here we give the formulae specialized to the case of the $^7\text{Li(n, } \gamma)^8\text{Li}$ and $^7\text{Be(p, } \gamma)^8\text{B}$ reactions, with E1 radiation. We include contributions only from neutron and proton channels respectively, and assume a common channel radius a for all channels. In the notation of Ref. [24], we therefore take $L = 1$, $l_f = l'_f = 1$, $\mu_e = \frac{7}{8}M_n$, $\bar{e}_1 = \pm \frac{3}{8}e$ (the upper and lower signs corresponding to the ^7Be and ^7Li reactions respectively), and also have $J_f = 1, 2$; $l_i = 0, 2$; $s_e = 1, 2$. Then

$$\sigma_{J_f} = \frac{\pi}{8k^2} \sum_{J_i} (2J_i + 1) \sum_{s_e l_i} |U_{s_e l_i, J_f}^J|^2, \quad (5)$$

with

$$\begin{aligned}
 U_{s_e l_i, J_f}^{J_i} = & -i e^{i(\omega_i - \phi_i)} \cdot 2 P_{l_i}^{1/2} k_\gamma^{3/2} \left[\sum_{\lambda \mu} A_{\lambda \mu}^{J_i} \gamma_{\lambda s_e l_i}^{J_i} \left(\gamma_{\mu}^{J_i} \gamma_{J_f}^{J_i}(\text{int}) \pm \frac{\sqrt{21}}{8} \frac{M_n^{1/2} e}{\hbar} a^2 N_f^{1/2} \right. \right. \\
 & \times \sum_c (-1)^{l/2} \gamma_{\mu c}^{J_i} \theta_{f \alpha s_1}^{J_i}(l 100 | 10) U(11 J_i s, l J_f) J_{1c}(l, 1) \left. \right) \\
 & \pm \frac{\sqrt{21}}{8} \frac{M_n^{1/2} e}{\hbar k} a N_f^{1/2} (-1)^{l/2} \theta_{f \alpha s_e s_1}^{J_i}(l_i 100 | 10) U(11 J_i s_e, l_i J_f) \\
 & \left. \times F_{l_i}(a) G_{l_i}(a) J_1'(l_i, 1) \right], \quad (6)
 \end{aligned}$$

where

$$\left[(A^{J_i})^{-1} \right]_{\lambda \mu} = (E_{\lambda}^{J_i} - E) \delta_{\lambda \mu} - \sum_c (S_l - B_l + i P_l) \gamma_{\lambda c}^{J_i} \gamma_{\mu c}^{J_i} \quad (c \equiv \alpha s l), \quad (7)$$

$$N_f^{-1} = 1 + \sum_{\alpha s} \frac{2(\theta_{f \alpha s_1}^{J_i})^2}{a} \int_a^\infty dr \left(\frac{W_{\alpha s_1}(r)}{W_{\alpha s_1}(a)} \right)^2, \quad (8)$$

and

$$\gamma_{\lambda c}^J = \theta_{\lambda c}^J \left(\frac{8}{7} \frac{\hbar^2}{M_n a^2} \right)^{1/2}. \quad (9)$$

For energies at which channel c is open, one has

$$J_{1c}(l, 1) = J_1''(l, 1) + i \frac{F_l(a) G_l(a)}{F_l^2(a) + G_l^2(a)} J_1'(l, 1), \quad (10)$$

where

$$J_1'(l, 1) = \frac{1}{a^2} \int_a^\infty dr r \frac{W_{\alpha s_1}(r)}{W_{\alpha s_1}(a)} \left(\frac{F_l(r)}{F_l(a)} - \frac{G_l(r)}{G_l(a)} \right) \quad (11)$$

and

$$J_1''(l, 1) = \frac{1}{a^2} \int_a^\infty dr r \frac{W_{\alpha s_1}(r)}{W_{\alpha s_1}(a)} \frac{F_l(a) F_l(r) + G_l(a) G_l(r)}{F_l^2(a) + G_l^2(a)}, \quad (12)$$

and for energies at which channel c is closed,

$$J_{1c}(l, 1) = \frac{1}{a^2} \int_a^\infty dr r \frac{W_{\alpha s_1}(r)}{W_{\alpha s_1}(a)} \frac{W_{\alpha s_1}(r)}{W_{\alpha s_1}(a)}. \quad (13)$$

For the ${}^7\text{Be}(p, \gamma){}^8\text{B}$ reaction, it is convenient to consider the astrophysical S factor rather than the cross section itself; these are related by

$$\sigma(E) = S(E) e^{-2\pi\eta/E}, \quad (14)$$

where η is the Sommerfeld parameter.

The ${}^7\text{Li} + n$ scattering lengths are given by [40]

$$a_{J_i} = a \left(1 - \sum_{\lambda\mu} A_{\lambda\mu}^{J_i}(E=0) \gamma_{\lambda J_i 0}^J \gamma_{\mu J_i 0}^J \right), \quad (15)$$

and the asymptotic normalization constants C_{I_s} of Refs. [19–21] by

$$C_{I_s} = (2/a)^{1/2} N_f^{1/2} \theta_{f\alpha_c s 1}^J [W_{\alpha_c s 1}(a)]^{-1}. \quad (16)$$

The measured widths of the 1^+ and 3^+ levels of ${}^8\text{Li}$ and ${}^8\text{B}$ are identified with R-matrix observed widths, given by

$$\Gamma^0(J) = \frac{2\Sigma_c (\gamma_{1c}^J)^2 P_1(E_{1c}^J)}{1 + \Sigma_c (\gamma_{1c}^J)^2 (dS_1/dE)_{E_{1c}^J}}, \quad (17)$$

provided that $B_c = S_1(E_{1c}^J)$.

The dimensionless reduced-width amplitudes θ are expressible in terms of spectroscopic amplitudes $\mathcal{S}^{1/2}$ and single-particle dimensionless reduced-width amplitudes θ_{sp} [41]:

$$\theta = \mathcal{S}^{1/2} \theta_{\text{sp}}, \quad (18)$$

where

$$\theta_{\text{sp}} = \left(\frac{a}{2} \right)^{1/2} \frac{u_c(a)}{\left[\int_0^a u_c^2(r) dr \right]^{1/2}} \quad (19)$$

and $r^{-1}u_c(r)$ is the radial wave function in channel c .

4. Procedure and results

In their fits to ${}^7\text{Li} + n$ data, Knox et al. [25] started with a shell-model calculation of the energies of ${}^8\text{Li}$ normal- and nonnormal-parity levels, and of the spectroscopic amplitudes of these levels for neutron and triton channels, using the PMM interaction. For each J^π value, the energies were identified with the R-matrix eigenenergies E_λ , and values of the R-matrix reduced-width amplitudes $\gamma_{\lambda c}$ were obtained from the spectroscopic amplitudes by using (for the neutron channels) Eqs. (9) and (18), with $a = 4.3$ fm and $\theta_{\text{sp}}^2 = 0.9$. The appropriate value of the boundary-condition parameter B_c was taken to be $B_c = S_c(E_\lambda)$, where λ_m labels the state for which $\gamma_{\lambda c}^2$ is largest. Before fitting the data, which covered the range $E_n = 0$ –8 MeV, Knox et al. converted these R-matrix parameter values to those corresponding to $B_c = S_c$ at an average neutron energy, using the formulae in Ref. [42]. Good fits to the data were then obtained by adjusting a few of these parameter values.

We modify this procedure in several respects. Since we are interested in low-energy data, for $E_n \leq 1$ MeV (and $E_p \leq 1$ MeV for ${}^7\text{Be}(p, \gamma){}^8\text{B}$), we omit the triton channels and include only the n_0 – n_3 channels with $l_{\text{max}} = 2$ (the n_2 and n_3 channels could also

be omitted without significantly affecting the results). Again we assume the PMM interaction in the shell-model calculation, which here makes use of the OXBASH shell-model code [43]. We calculate energies, spectroscopic amplitudes and also E1-transition matrix elements. The internal contributions to the photon reduced-width amplitudes (see Eq. (6)) are calculated from the E1-transition matrix elements, using

$$\gamma_{\lambda\gamma J_f}^{J_i}(\text{int}) = [4\pi/27(2J_f + 1)]^{1/2} \text{RME}(\text{DT} = 1). \quad (20)$$

As in Knox et al. [25], we include four states for each of $J^\pi = 1^+, 2^+, 3^+, 1^-$, and 2^- , and three states for $J^\pi = 3^-$. These R-matrix parameter values are assumed to correspond to the values $B_c = S_c(\bar{E}_c)$, where $\bar{E}_c$ is defined by [44]

$$\bar{E}_c = \frac{\sum_\lambda E_\lambda \gamma_{\lambda c}^2}{\sum_\lambda \gamma_{\lambda c}^2}. \quad (21)$$

For each J^π , we assume that the separations of the shell-model energies are the separations of the eigenenergies E_λ . Since the formulae of Section 3 implicitly assume a one-level approximation for the normal-parity levels, we determine the zero of the eigenenergy scale for each J^+ by fitting the observed energies $E_1^{J^+}$ of the low-lying 2^+ , 1^+ , and 3^+ levels, with parameter values converted [42] to $B_c = S_c(E_1^{J^+})$. For the nonnormal-parity levels, the parameter values are converted [42] to those for $B_c = S_c(E_x = 3 \text{ MeV})$, as this is approximately the energy of the lowest 1^- and 2^- states, which give the main contributions to the low-energy ${}^7\text{Li}(n, \gamma){}^8\text{Li}$ cross section. Changing this energy from 3 to 4 MeV does not significantly change the results.

In order to avoid a complicated iterative procedure, these initial parts of the calculation are done with the fixed value $\theta_{\text{sp}}^2 = 0.6$ [41] and with $a = 1.45(7^{1/3} + 1) \text{ fm} = 4.22 \text{ fm}$ [41]. In the later stages of fitting the ${}^7\text{Li} + n$ data and calculating the ${}^7\text{Be}(p, \gamma){}^8\text{B}$ cross section, where differences between ${}^7\text{Li} + n$ and ${}^7\text{Be} + p$ channels are presumably significant, it is assumed that the spectroscopic amplitudes are the same in the mirror systems and that θ_{sp} is obtained from Eq. (19), using $u_c(r)$ calculated for a central Woods–Saxon potential (cut-off at $r = a$), with conventional radius $1.25 \times 7^{1/3} \text{ fm} = 2.39 \text{ fm}$ and diffuseness 0.65 fm, and with the depth adjusted to fit the binding energy or resonance energy appropriate to the state and channel involved. Also the internal contributions to the photon reduced-width amplitudes are assumed to be equal and opposite in the mirror systems. In order to determine the sensitivity of the results to the choice of channel radius, this part of the calculation is done for $a = 4, 5$, and 6 fm.

Fits to the ${}^7\text{Li} + n$ data given in Section 2 are made by varying some of the parameter values. The zeros of the 1^- and 2^- energy scales are adjusted in order to fit the scattering lengths given in Eq. (1). Because contributions from 3^- levels are due to d-wave neutrons and are small, we do not adjust the 3^- energy scale. A comparison of predicted values of the other ${}^7\text{Li} + n$ quantities with the experimental values is given in Table 2. There is reasonable agreement, especially for $a = 4 \text{ fm}$. We obtain better agreement by varying the parameters to which these quantities are most sensitive. These are the spectroscopic amplitudes $(\mathcal{S}_{f\alpha_e s_e}^{J_f})^{1/2}$ for the ${}^7\text{Li}(\text{g.s.}) + n_0$ channels, with $J_f = 1, 2$ and with $s_e = J_i = 1, 2$, and the internal contributions to the photon reduced-

Table 2
Comparison of shell-model predicted values with experimental values

	a (fm)	$\sigma_{n\gamma}(E_{th})$ (mb)	$\mathcal{R}$	W_+	$\mathcal{P}(2^+)$	$\mathcal{P}(1^+)$
Predicted	4	41.2	0.119	0.864	1.034	0.450
	5	35.7	0.140	0.834	1.034	0.450
	6	27.2	0.172	0.794	1.034	0.450
Experimental		45.4 ± 3.0	0.106 ± 0.010	> 0.86	0.87 ± 0.174	0.48 ± 0.096

width amplitudes $\gamma_{\lambda\gamma J_f}^{J_i}(\text{int})$ for $J_i = 1, 2$, $J_f = 1, 2$, and $\lambda = 1$. There are eight of these parameters, while the data to be fitted are four values and one lower limit. Thus we vary at most four parameters at a time. Of the 70 possible sets with four parameters varied, only ten give exact fits to the data with W_+ in the required range. Table 3 gives these for the case $a = 4$ fm. All these fits (and the corresponding ones for $a = 5$ and 6 fm) give $\sigma_{n\gamma_0}(E_n = 30 \text{ keV}) = 35.6\text{--}36.0 \text{ } \mu\text{b}$, in good agreement with the measured value of Nagai et al. [34]. The predicted values of $\sigma_{n\gamma}$ for higher values of E_n do not vary much between the different fits; for example, $\sigma_{n\gamma}(E_n = 600 \text{ keV})$ lies between 4.5 and 5.5 μb for $a = 4, 5$, and 6 fm. Fig. 1 shows the calculated $\sigma_{n\gamma}$, for the first row of modified values in Table 3, compared with experimental values.

Table 3 also gives the calculated values of $S_{17}(0)$ for the ${}^7\text{Be}(\text{p}, \gamma){}^8\text{B}$ reaction. It is seen that these are the same for each of the parameter sets that gives an exact fit to the ${}^7\text{Li} + n$ data, namely $S_{17}(0) = 16.2 \text{ eV} \cdot \text{b}$. For $a = 5$ and 6 fm, the corresponding values are 18.4 and 19.0 $\text{eV} \cdot \text{b}$. From Eq. (16) and $|G_{1s}|^2 = (\pi/\mu^2)|C_{1s}|^2$ [20,21], we find $|G_{11}|^2 + |G_{12}|^2 = 0.079 \text{ fm}$ for each of the modified-parameter sets in Table 3, and similarly 0.089 and 0.093 fm for $a = 5$ and 6 fm respectively. These are comparable

Table 3
Parameter values giving exact fit to ${}^7\text{Li} + n$ data for $a = 4$ fm

$(\mathcal{P}_{1\alpha_e J_{i1}}^{J_i})^{1/2}$				$\gamma_{1\gamma J_f}^{J_i}(\text{int}) (\text{MeV}^{1/2} \text{fm}^{3/2})$				W_+	$S_{17}(0)$ (eV·b)
J_i	1	2	2	1	1	2	2		
J_f	1	2	1	2	1	2	1	2	
<i>Original values</i>									
	−0.411	0.458	−0.530	0.908	0.015	0.162	0.128	−0.199	0.864 18.4
<i>Modified values</i>									
	−0.470	0.215	−0.509					−0.044	0.947 16.2
	−0.446	0.215			−0.266			−0.044	0.954 16.2
	−0.446	0.215					0.095	−0.044	0.950 16.2
	−0.470		−0.509	0.812				0.056	0.868 16.2
	−0.446			0.812	−0.266			0.056	0.874 16.2
	−0.446			0.812			0.095	0.056	0.870 16.2
		0.215	−0.558		−0.802			−0.044	0.962 16.2
		0.215	−0.558				0.054	−0.044	0.953 16.2
			−0.558	0.812	−0.802			0.056	0.883 16.2
			−0.558	0.812			0.054	0.056	0.873 16.2

Table 4

Values of $\mathcal{S}(3^+)$ obtained from fitting widths of 3^+ levels of ^8Li and ^8B and from stripping [35]

a (fm)	$\mathcal{S}(3^+)$		
	^8Li	^8B	Stripping
4	0.36 ± 0.08	0.30 ± 0.03	
5	0.46 ± 0.10	0.28 ± 0.03	0.25 ± 0.06
6	0.65 ± 0.13	0.29 ± 0.03	

with the previous values of 0.077 fm [20] and 0.082 fm [21]. The width of the 1^+ level of ^8B may be calculated from Eq. (17). The same value is obtained for each of the modified parameter sets, and is 29.9, 34.0, and 35.7 keV for $a = 4, 5$, and 6 fm respectively.

So far the fits have taken no account of the uncertainties in the experimental values of the $^7\text{Li} + n$ data, as given in the bottom row of Table 2. Using the first set of modified parameters in Table 3, we find that fits to data in which $\sigma_{n\gamma}(E_{\text{th}})$ is varied through the allowable range (42.4–48.4 mb) all give the same values of $S_{17}(0)$ and of $\Gamma(^8\text{B}, 1^+)$, although $\sigma_{n\gamma_0}(E_n = 30 \text{ keV})$ varies from 33.3 to 38.1 μb . Similar variations in the fitted value of $\mathcal{R}$ produce no significant changes in other quantities. Varying $\mathcal{S}(2^+)$ affects only $S_{17}(0)$, which ranges from 15.6 to 18.5 eV $\cdot$ b, while variations of $\mathcal{S}(1^+)$ give $\Gamma(^8\text{B}, 1^+)$ values from 25 to 35 keV.

The $^7\text{Be}(p, \gamma)^8\text{B}$ S factor at higher energies, calculated using the first row of modified-parameter values in Table 3, and the corresponding values for $a = 5$ and 6 fm, are shown in Fig. 2 in comparison with the experimental data (with normalization and error bars as in Ref. [4]).

Although not directly relevant to the low-energy $^7\text{Be}(p, \gamma)^8\text{B}$ cross section, a test of the present approach may be made by fitting the observed widths of the 3^+ levels of ^8Li and ^8B , as given in Table 1, and comparing the required spectroscopic factors with each other and with that obtained from the stripping reaction $^7\text{Li}(d, p)^8\text{Li}$. The only available stripping value is that of Macfarlane and French [35]; they agreed well with Schiffer et al. [36] in values for the 2^+ and 1^+ levels (see Table 1). Fitting the widths using Eq. (17), together with Eqs. (9) and (18), with only the value of $\mathcal{S}_{1\alpha, 21}^{3^+} \equiv \mathcal{S}(3^+)$ adjusted, leads to the results in Table 4. The stripping value includes the estimated uncertainty of 25% [35]. There is reasonable agreement between the various values, especially for $a = 4$ fm.

5. Discussion

In our R-matrix fits to the $^7\text{Li} + n$ data with $a = 4, 5$, and 6 fm, the changes required in the shell-model parameter values are smaller for the smaller channel radii. Also values of $\mathcal{S}(3^+)$ from various sources give a preference for a small value of a . On the other hand there is best agreement with the experimental value of $\Gamma(^8\text{B}, 1^+)$ for $a = 6$

fm, but an acceptable fit can be obtained for $a = 4$ fm if $\mathcal{S}(1^+)$ is increased within its allowed range. The fits lead to $S_{17}(0) = 16, 18,$ and 19 eV $\cdot$ b for $a = 4, 5,$ and 6 fm respectively. With allowance for uncertainties in the data, we suggest a best value $S_{17} = 17 \pm 3$ eV $\cdot$ b.

We now comment on some of the other methods for obtaining values of $S_{17}(0)$, which were mentioned in the introduction. Kim et al. [5] defined an alternative S factor, $\tilde{S}(E)$, by replacing $e^{-2\pi\eta}$ in Eq. (14) by a transmission coefficient, and they assumed that $\tilde{S}(E)$ could be represented by a quadratic function of E over the range of the data fitted ($E \leq 2700$ keV, excluding the resonance region). Extrapolation to low energies then gave smaller cross sections than those obtained using calculated energy dependences. The small values are, however, due more to the assumption of a quadratic energy dependence than to the replacement of S by $\tilde{S}$ (which caused less than 10% reduction). All model calculations [9–11, 14–18] give a sharp upturn of $S(E)$ as $E \rightarrow 0$, and semianalytic calculations [49] have shown that $S(E)$ can be well represented by a quadratic function of E only for $E \leq 100$ keV, due to the small binding energy of the final state.

The present R-matrix approach takes account of channels involving the $\frac{1}{2}^-$ first-excited state of ${}^7\text{Be}$ (and higher states). Johnson et al. [17] criticized the potential approach [9,10], claiming that complex potentials should be used for $E > 430$ keV, due to the opening of the ${}^7\text{Be}(\frac{1}{2}^-)$ channel. The only supporting evidence that they gave was that, for the mirror system, the ${}^7\text{Li}(n, n'){}^7\text{Li}(\frac{1}{2}^-)$ cross section shows an abrupt rise from threshold. This indicates significant s-wave n' , and therefore $J^\pi = 0^-$ or 1^- . These J^π values, however, contribute less than 7% of the low-energy ($E_p \leq 1$ MeV) ${}^7\text{Be}(p, \gamma){}^8\text{B}$ cross section, which is dominated by 2^- contributions, and the evidence does not suggest a complex potential for this J^π .

In contrast with the R-matrix approach, which has flexibility to fit experimental data due to the presence of adjustable energy-independent parameters, the cluster-model calculations [14–18] contain no adjustable parameters, once the NN interaction is chosen, and more tolerance is shown in requiring agreement of the cluster-model predictions with experimental data. Only in Refs. [14,17] was the model used in the mirror ${}^7\text{Li} + n$ system to calculate values of the thermal-neutron properties given in Eqs. (1)–(4). Descouvemont and Baye [14] found $a_1 = -0.72$ fm, $a_2 = -3.26$ fm, $\sigma_{n\gamma}(E_{\text{th}}) = 54.8$ mb, $\mathcal{R} = 0.111$, and $W_+ = 0.79$; Johnson et al. [17] gave $a_1 = 1.25$ fm, $a_2 = 0.26$ fm, and apparently $\sigma_{n\gamma}(E_{\text{th}}) = 30.6$ mb. They argued, however, that the accuracy of their ${}^7\text{Li}(n, \gamma){}^8\text{Li}$ calculations does not indicate the accuracy of their calculated ${}^7\text{Be}(p, \gamma){}^8\text{B}$ cross section, because the former are sensitive to internal contributions while the latter is not. All the cluster-model calculations assume a ${}^3\text{He} + \alpha$ structure for the $\frac{3}{2}^-$ and $\frac{1}{2}^-$ states of ${}^7\text{Be}$. It has been shown, however, that calculations based on an $\alpha + t$ structure for the $\frac{3}{2}^-$ and $\frac{1}{2}^-$ states of the mirror nucleus ${}^7\text{Li}$ are not in agreement with experimental values for the ground-state quadrupole moment and the $B(E2)$ connecting the two states [50]. It may be this assumption that leads to the “high” values of $S_{17}(0)$ from the cluster-model calculations.

Recent standard-solar-model calculations [3] have adopted the value $S_{17}(0) = 22.4 \pm$

2.1 eV · b, which was taken from a cluster-model calculation [17]. The value $S_{17}(0) = 17 \pm 3$ eV · b suggested here is more than 20% lower, and is consistent with the results of many other studies [4,6,10,21,22]. Its use in a solar-model calculation would lead to a significant reduction in the calculated flux of ^8B neutrinos, so reducing the severity of the solar-neutrino problem.

References

- [1] J.N. Bahcall, Neutrino astrophysics (Cambridge Univ. Press, Cambridge, 1989).
- [2] J.N. Bahcall, N.A. Bahcall and G. Shaviv, Phys. Rev. Lett. 20 (1968) 1209.
- [3] J.N. Bahcall and M.H. Pinsonneault, Rev. Mod. Phys. 64 (1992) 885.
- [4] F.C. Barker and R.H. Spear, Ap. J. 307 (1986) 847.
- [5] Y.E. Kim, M. Rabinowitz, J.H. Yoon and R.A. Rice, Int. J. Theor. Phys. 32 (1993) 301.
- [6] T. Motobayashi et al., Phys. Rev. Lett. 73 (1994) 2680.
- [7] K. Langanke and T.D. Shoppa, Phys. Rev. C 49 (1994) R1771.
- [8] M. Gai and C.A. Bertulani, Comment on E2 contribution to the $^8\text{B} \rightarrow \text{p} + ^7\text{Be}$ Coulomb dissociation cross section, preprint UConn-40870-0005 (1994).
- [9] T.A. Tombrello, Nucl. Phys. 71 (1965) 459.
- [10] F.C. Barker, Aust. J. Phys. 33 (1980) 177.
- [11] K.H. Kim, M.H. Park and B.T. Kim, Phys. Rev. C 35 (1987) 363.
- [12] F.C. Barker, Phys. Rev. C 37 (1988) 2920.
- [13] H. Krauss, K. Grün, T. Rauscher and H. Oberhummer, Ann. der Phys. 2 (1993) 258.
- [14] P. Descouvemont and D. Baye, Nucl. Phys. A 487 (1988) 420.
- [15] P. Descouvemont and D. Baye, Nucl. Phys. A 567 (1994) 341.
- [16] E. Kolbe, K. Langanke and H.J. Assenbaum, Phys. Lett. B 214 (1988) 169.
- [17] C.W. Johnson, E. Kolbe, S.E. Koonin and K. Langanke, Ap. J. 392 (1992) 320.
- [18] A. Csótó, K. Langanke, S.E. Koonin and T.D. Shoppa, The $^7\text{Be}(\text{p}, \gamma)^8\text{B}$ cross section and the properties of ^7Be , preprint (1994).
- [19] A.M. Mukhamedzhanov and N.K. Timofeyuk, JETP Lett. 51 (1990) 282.
- [20] A.M. Mukhamedzhanov and N.K. Timofeyuk, Sov. J. Nucl. Phys. 51 (1990) 431.
- [21] H.M. Xu, C.A. Gagliardi, R.E. Tribble, A.M. Mukhamedzhanov and N.K. Timofeyuk, The overall normalization of the astrophysical S factor and the nuclear vertex constant for $^7\text{Be}(\text{p}, \gamma)^8\text{B}$ reactions, Texas A&M preprint 94-09 (1994).
- [22] K. Riisager and A.S. Jensen, Phys. Lett. B 301 (1993) 6.
- [23] A.M. Lane and R.G. Thomas, Rev. Mod. Phys. 30 (1958) 257.
- [24] F.C. Barker and T. Kajino, Aust. J. Phys. 44 (1991) 369.
- [25] H.D. Knox, D.A. Resler and R.O. Lane, Nucl. Phys. A 466 (1987) 245.
- [26] L. Koester, K. Knopf and W. Waschkowski, Z. Phys. A 312 (1983) 81.
- [27] J.E. Lynn, E.T. Jurney and S. Raman, Phys. Rev. C 44 (1991) 764.
- [28] A.D. Gul'ko, S.S. Trostin and A. Hudoklin, Sov. J. Nucl. Phys. 6 (1968) 477.
- [29] D.J. Hughes, D. Hall, C. Eggler and E. Goldfarb, Phys. Rev. 72 (1947) 646.
- [30] E.A. Koltypin and V.M. Morozov, Sov. Phys. Dokl. 1 (1956) 655.
- [31] W.L. Imhof, R.G. Johnson, F.J. Vaughn and M. Walt, Phys. Rev. 114 (1959) 1037.
- [32] von L. Jarczyk, J. Lang, R. Müller and W. Wölfl, Helv. Phys. Acta 34 (1961) 483.
- [33] M. Wiescher, R. Steininger and F. Käppeler, Ap. J. 344 (1989) 464.
- [34] Y. Nagai, M. Igashira, N. Mukai, T. Ohsaki, F. Uesawa, K. Takeda, T. Ando, H. Kitazawa, S. Kubono and T. Fukuda, Ap. J. 381 (1991) 444.
- [35] M.H. Macfarlane and J.B. French, Rev. Mod. Phys. 32 (1960) 567.
- [36] J.P. Schiffer, G.C. Morrison, R.H. Siemssen and B. Zeidman, Phys. Rev. 164 (1967) 1274.
- [37] B.W. Filippone, A.J. Elwyn, C.N. Davids and D.D. Koetke, Phys. Rev. C 28 (1983) 2222.
- [38] F. Ajzenberg-Selove, Nucl. Phys. A 490 (1988) 1.
- [39] R.L. McGrath, J. Cerny and E. Norbeck, Phys. Rev. Lett. 19 (1967) 1442.

- [40] F.C. Barker, Aust. J. Phys. 30 (1977) 113.
- [41] A.M. Lane, Rev. Mod. Phys. 32 (1960) 519.
- [42] F.C. Barker, Aust. J. Phys. 25 (1972) 341.
- [43] B.A. Brown, A. Etchegoyan and W.D.M. Rae, OXBASH – the Oxford–Buenos Aires – MSU shell model code, Michigan State University Cyclotron Laboratory Internal report MSUCL-524 (1986).
- [44] F.C. Barker and E.K. Warburton, Nucl. Phys. A 487 (1988) 269.
- [45] R.W. Kavanagh, Nucl. Phys. 15 (1960) 411.
- [46] P.D. Parker, Ap. J. (Lett.) 153 (1968) L85.
- [47] R.W. Kavanagh, T.A. Tombrello, J.M. Mosher and D.R. Goosman, Bull. Am. Phys. Soc. 14 (1969) 1209.
- [48] F.J. Vaughn, R.A. Chalmers, D. Kohler and L.F. Chase, Phys. Rev. C 2 (1970) 1657.
- [49] R.D. Williams and S.E. Koonin, Phys. Rev. C 23 (1981) 2773.
- [50] W.J. Vermeer, R.H. Spear and F.C. Barker, Nucl. Phys. A 500 (1989) 212;
F.C. Barker, Y. Kondō and R.H. Spear, Aust. J. Phys. 42 (1989) 597.